Acceptance Test Report

Face Detection Game - Test Cases & Results

Embedded Systems - Face Detection Game Project

SDU - Software Engineering

May 2026

Executive Summary

This report documents comprehensive acceptance testing of the Face Detection Game application. The testing encompasses face detection accuracy, game mechanics, performance, and user interaction. All critical game mechanics have been tested and validated. Three significant issues were identified during testing that impact user experience and require attention.

Testing Scope

Testing Targets

- Face detection accuracy and reliability
- Game mechanics (scoring, strikes, difficulty scaling)
- Performance metrics (frame rate, responsiveness)
- Face recognition requirements and angle tolerance
- User input handling (button responsiveness)
- Game state management (IDLE, RUNNING, PAUSED, END)
- Real-time HUD updates and feedback

Acceptance Test Cases

Category 1: Face Detection

Tests verify that the system can detect single and multiple faces in various scenarios.

#	Test Case	Expected Result	
4	Bounding Boxes	Green bounding boxes render around detected	PASS
5	Empty Frame	System reports 'No faces detected' on blank	PASS
6	Different Face Sizes	Detection works with faces at various distances	PASS

Category 2: Game Mechanics

Tests verify core game rules: scoring, strikes, game state transitions, and difficulty scaling.

#	Test Case	Expected Result	
6	Minimum Timer	Time limit minimum is 0.5 seconds	PASS

Category 3: User Interface & Controls

Tests verify HUD display, state transitions, and keyboard controls.

#	Test Case	Expected Result	
			PASS
2	RUNNING State	Timer active, score/strikes displayed, target	PASS
8	HUD Updates	Real-time display of FPS, faces, confidence, score	PASS

Category 4: Performance

Tests measure system responsiveness and real-time performance.

#	Test Case	Expected Result	
4	Memory Usage	No memory leaks during extended play (>5 min)	PASS

Category 5: Face Recognition Requirements

Tests verify face recognizability under various conditions.

#	Test Case	Expected Result	
6	Lighting Variation	Detection works in varying light conditions	PASS

Test Summary

Results Overview

Total Tests Executed: 29
Passed: 24 (82.8%)
Failed: 5 (17.2%)
Status: CONDITIONAL PASS - Core functionality works; Performance and angle tolerance issues require resolution.

Identified Issues & Improvements

The following issues were identified during acceptance testing. These impact user experience and should be prioritized for resolution.

High Priority Issues

HIGH **Frame Rate Performance (Very Slow)**

Description: The application experiences low frame rates during gameplay, achieving only 15-20 FPS instead of the target 30+ FPS. This causes jerky camera updates and delayed face detection, degrading the user experience.
Impact: Makes gameplay difficult and frustrating; delays feedback on face detection.
Recommended Fix: Optimize face detection algorithm (consider multi-threading, reduce image resolution for detection, or use lighter DNN model).

HIGH **Face Recognizability - Limited Angle Tolerance**

Description: The face detection system requires faces to be positioned nearly straight toward the camera (0° angle). Faces at even slight angles (15-45°) are often not detected. This severely limits user flexibility and makes gameplay difficult, especially in multiplayer scenarios.
Impact: Unnatural gameplay experience; restricts valid player positions; fails in varied angles.
Recommended Fix: Use 3D face detection algorithms or multi-angle cascade classifiers; train on diverse face angles; enable profile face detection.

HIGH **Button Input Responsiveness - Hold Requirement**

Description: Keyboard controls (especially SPACE, A, D for pause and target adjustment) require continuous holding rather than simple presses. This makes control feel unresponsive and requires unnatural input patterns.
Impact: Poor user interaction; unintuitive control scheme; frustrating gameplay experience.
Recommended Fix: Implement proper key press/release event handling; use debouncing for toggle functions (SPACE for pause should work on single press, not hold).

Recommendations & Next Steps

Priority 1: Critical Path (Address Before Release)

- Resolve frame rate performance - target 30 FPS minimum
- Improve face angle tolerance to at least ±30 degrees
- Fix button input handling - change from 'hold' to 'press' semantics

Priority 2: Enhancements (Post-Release)

- Add configurable sensitivity settings for face detection
- Implement adaptive difficulty based on detected face angles
- Add visual feedback for why faces are not detected (lighting, angle, distance)
- Implement input remapping and alternative control schemes

Priority 3: Testing

- Conduct usability testing with diverse user groups
- Test on various camera hardware (integrated vs external cameras)
- Performance testing on lower-end hardware specifications
- Long-duration stress testing (1+ hour gameplay sessions)

Conclusion

The Face Detection Game demonstrates solid core functionality with successful face detection, game mechanics, and state management. However, three significant issues limit the user experience: low frame rates, limited angle tolerance for face recognition, and unintuitive button input handling.

The application is currently in a playable state but falls short of professional polish. Addressing the identified high-priority issues is essential before production release. Once these issues are resolved, the game should provide an engaging and responsive user experience.

Recommended Action: Prioritize resolution of the three high-impact issues identified in the 'Identified Issues & Improvements' section. Following resolution, conduct a follow-up acceptance test to verify corrections.

Test Report Metadata

Report Date: May 05, 2026
Project: Embedded Systems - Face Detection Game
Testing Team: Quality Assurance
Test Environment: Windows 10/11, Python 3.8+, OpenCV 4.5+
Report Version: 1.0